Chapter – 1
Fundamental of Microprocessor

Introduction

- A Microprocessor is a multipurpose programmable, clock driven, register based electronic device.
- It reads binary instructions from a memory, accepts binary data as input, processes them according to those instructions & provide results as output.
- The microprocessor operates in binary 0 and 1 known as bits are represented in terms of electrical voltages in the machine that means 0 represents low voltage level and 1 represents high voltage level.
- Each microprocessor recognizes and processes a group of bits called the word, microprocessors are classified according to their word length such as 8 bits microprocessor with 8 bit word and 32 bit microprocessor with 32 bit word

Basic Block Diagram of a Computer

- Traditionally, the computer is represented with four components such as memory, input, output and central processing unit (CPU) which consists of arithmetic logic unit (ALU) and control unit (CU).
- The CPU contains various registers to store data, the ALU to perform arithmetic and logical operations, instruction decoders, counters and control lines.

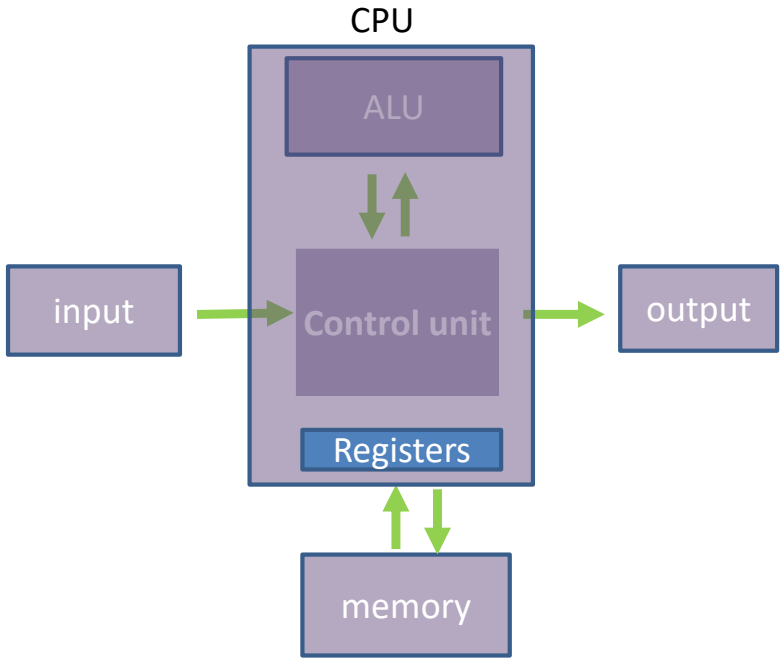

Fig : Traditional Block diagram of a computer

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- Later on , traditional block diagram was replaced with computer having microprocessor as CPU which is known as microcomputer.
- Here CPU was designed using integrated circuit technology (IC's) which provided the possibility to build the CPU on a single chip.

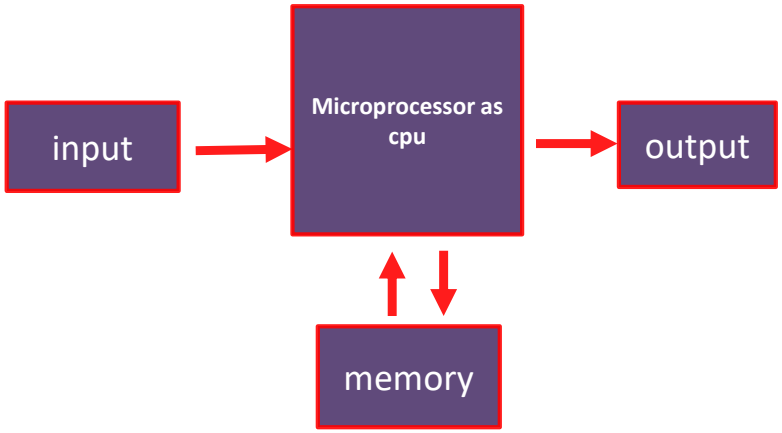

Fig : Block Diagram of a computer with the Microprocessor as CPU

Organization of a microprocessor based system

- Microprocessor based system includes three components microprocessor, input/output and memory (read only and read/write).
- These components are organized around a common communication path called a bus.

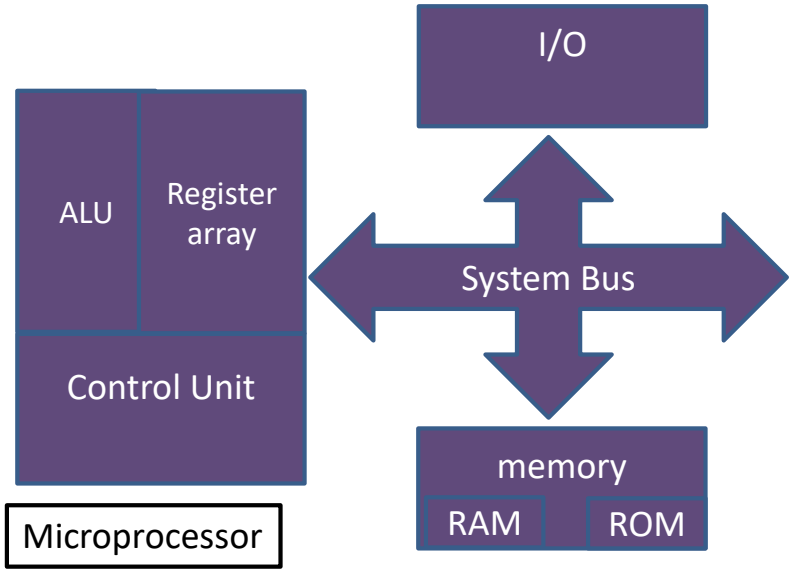

Fig 1.3: Microprocessor Based System with Bus Architecture

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Microprocessor:

- It is clock driven semiconductor device consisting of electronic logic circuits manufactured by using either a large scale integration (LSI) or very large scale integration (VLSI) technique.
- It is capable of performing various computing functions and making decisions to change the sequence of program execution. It can be divided in to three segments.

  A. Arithmetic/Logic unit:

  B. Register Array

  C. Control Unit

Cont …

A. **Arithmetic/Logic unit:** It performs arithmetic operations as addition and subtraction and logic operations as AND, OR & XOR.
B. **Register Array:** The registers are primarily used to store data temporarily during the execution of a program and are accessible to the user through instruction. The registers can be identified by letters such as B, C, D, E, H and L.
C. **Control Unit:** It provides the necessary timing and control signals to all the operations in the microcomputer. It controls the flow of data between the microprocessor and memory & peripherals.

Cont…

Memory:

- Memory stores binary information such as instructions and data, and provides that information to the microprocessor whenever necessary.
- To execute programs, the microprocessor reads instructions and data from memory and performs the computing operations in its ALU.
- Results are either transferred to the output section for display or stored in memory for later use. Memory has two sections.

  A. Read only Memory (ROM): Used to store programs that do not need alterations and can only read.

  B. Read/Write Memory (RAM): Also known as user memory which is used to store user programs and data. The information stored in this memory can be easily read and altered.

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Input/Output:

- It communicates with the outside world using two devices input and output which are also Known as peripherals.
- The input device such as keyboard, switches, and analog to digital converter transfer binary information from outside world to the microprocessor.
- The output devices transfer data from the microprocessor to the outside world. They include the devices such as LED, CRT, digital to analog converter, printer etc.

System Bus:

- It is a communication path between the microprocessor and peripherals; it is nothing but a group of wires to carry bits.

Bus organization

- Bus is a common channel through which bits from any sources can be transferred to the destination.
- A typical digital computer has many registers and paths must be provided to transfer instructions from one register to another.
- The number of wires will be excessive if separate lines are used between each register and all other registers in the system.

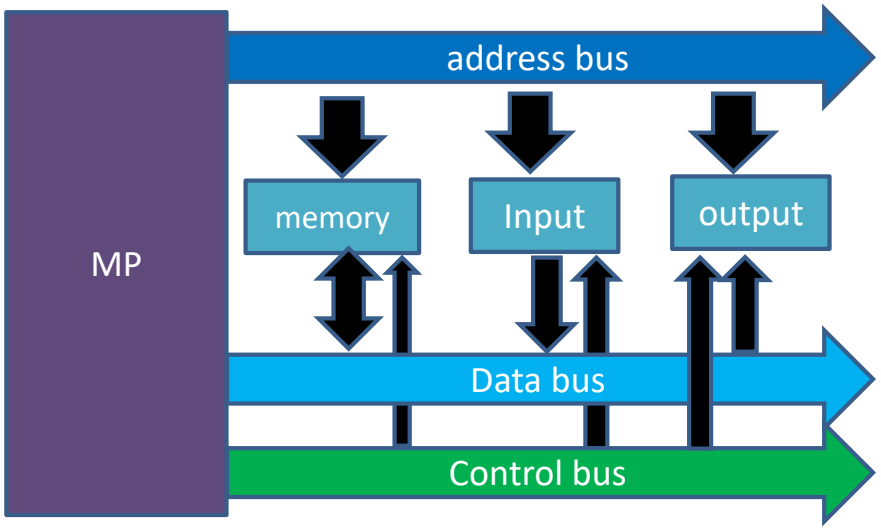

Fig: Bus Organization

Cont…

- Bus organisation are Classified into three functional groups; data, address and control lines.

a. Address Bus:

- The address bus is used by the CPU to send the address of the memory location or the input/output port that is to be accessed at the instant.
- It is a unidirectional bus i.e. the address can be transferred in one direction only and that is from CPU to the required port or location.
- Whether it is a read operation or write operation the CPU calculates the address of the required data and sends it on the data bus for the execution of the required operation. The maximum number of memory locations that can be accessed in a system is determined by the number of lines of an address bus.

b. Data bus

- A data bus is used to carry the data and instructions from the CPU to memory and peripheral devices and vice versa. Thus it is a bidirectional bus.
- It is one of most important parts of the connections to the CPU because every program instruction and every byte of data must travel across the bus at some point.
- The size of the data bus is measured in bits.
- Generally, a microprocessor is called n-bit processors. Thus as the CPU became more advanced, the data bus grew in size. A 64-bit data bus can transfer 8 bytes in every bus cycle

C. Control Bus

- A control bus contains various individual lines carrying synchronizing signals that are used to control. Various peripheral devices connected to the CPU.
- The common signals that are transferred to the control bus from CPU to devices and vice versa are memory read, memory writes, I/O read, I/O write etc.

Stored Program Concept and Von-Neumann Machine

- The simplest way to organize a computer is to have one processor, register and instruction code format with two parts op-code and address/operand.
- The memory address tells the control where to find an operand in memory.
- This operand is read from memory and used as data to be operated on together with the data stored in the processor register.
- Instructions are stored in one section of same memory.
- It is called as stored program concept.

Conti…

- It could be facilitated if the program could be represented in a form suitable for storing in memory alongside the data.
- So the computer could get its instructions by reading from the memory and program could be set or altered by setting the values of a portion of memory.
- This approach is known as 'stored- program concept' was first adopted by John Von Neumann and such architecture is named as von-Neumann architecture

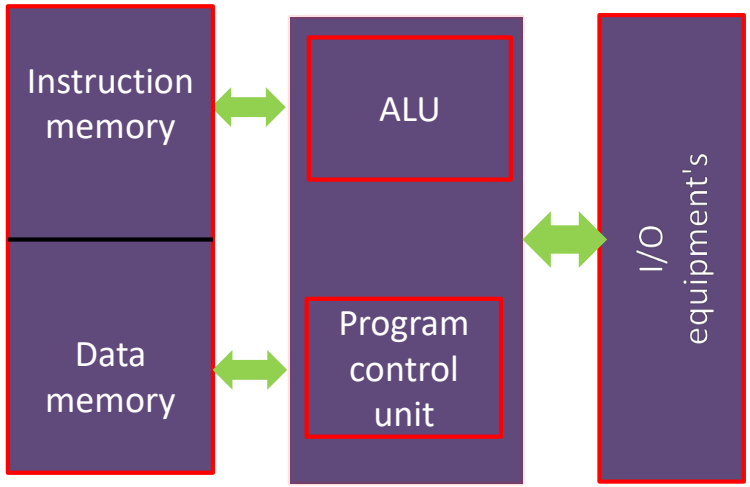

Fig: Von –Neumann Architecture

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- The Von–Neumann architecture is the fundamental basis for the architecture of modern digital computers.
- The main memory is used to share both data and instructions.
- The arithmetic and logic unit is capable of performing arithmetic and logical operation on binary data.
- The program control unit interprets the instruction in memory and causes them to be executed.
- The I/O unit gets operated from the control unit.
- The storage location of control unit and ALU are called registers and the various models of registers are:

Conti…

- **MAR** – memory address register – contains the address in memory of the word to be written into or read from MBR.
- **MBR** – memory buffer register – consists of a word to be stored in or received from memory.
- **IR –** instruction register – contains the 8-bit op-code instruction to be executed.
- **IBR** – instruction buffer register – used to temporarily hold the instruction from a word in memory.
- **PC** - program counter - contains the address of the next instruction to be fetched from memory.
- **AC & MQ** (Accumulator and Multiplier Quotient) - holds the operands and results of ALU after processing.

Harvard Architecture

- In von-Neumann architecture, the same memory is used for storing instructions and data.
- Similarly, a single bus called data bus or address bus is used for reading data and instructions from or writing to memory.
- It had limited the processing speed for computers.
- But the Harvard architecture based computer consists of separate memory spaces for the programs (instructions) and data.
- Each space has its own address and data buses.
- So instructions and data can be fetched from memory concurrently and provides significance processing speed improvement.
- There are two data and two address buses multiplexed for data bus and address bus.
- There are two blocks of RAM chips one for program memory and another for data memory addresses.

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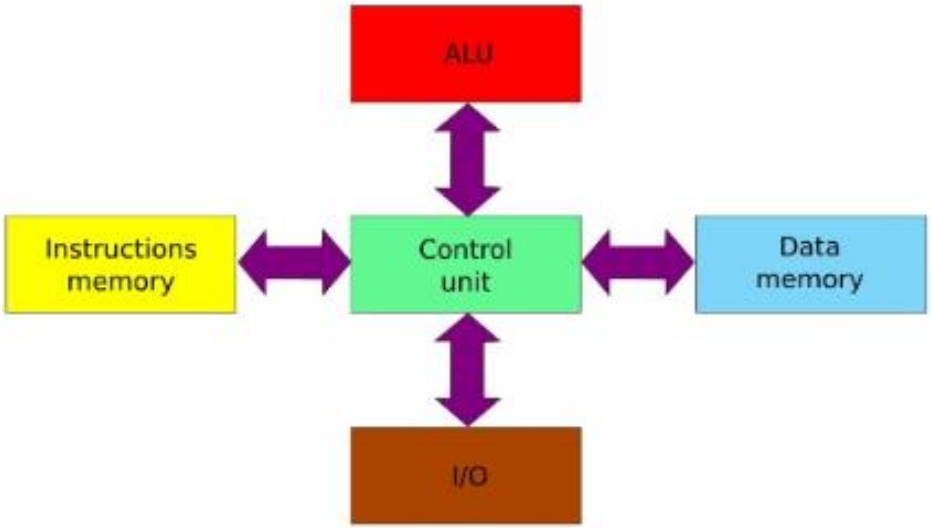


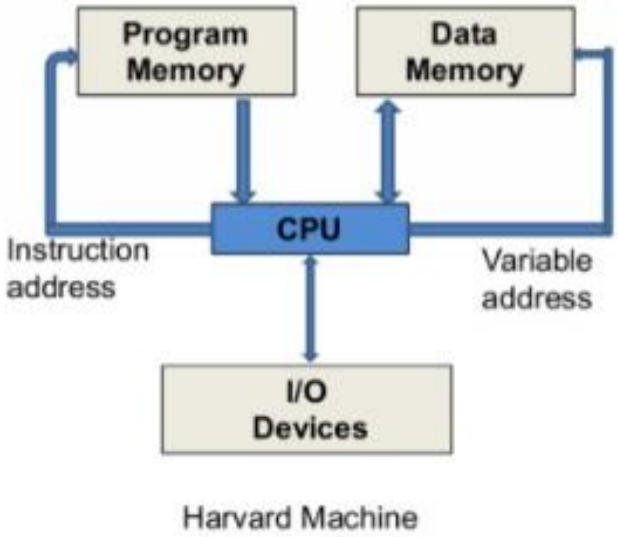


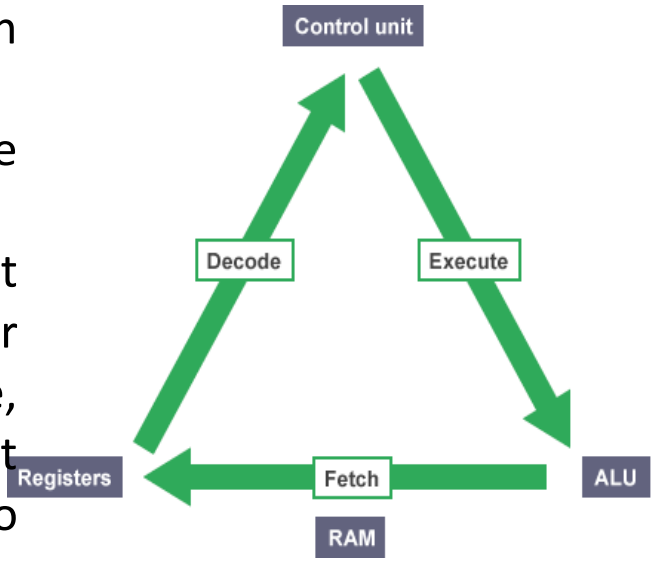
Fig: Harvard Architecture Based Microprocessor

Fetch-decode-execute cycle

- The main job of the **CPU** is to **execute** programs using the **fetch-decode-execute cycle** (also known as the **instruction cycle**). This cycle begins as soon as you turn on a computer.
- To execute a program, the program code is copied from secondary storage into the main memory. The CPU's **program counter** is set to the memory location where the first instruction in the program has been stored, and execution begins. The program is now running.
- In a program, each **machine code** instruction takes up a slot in the main memory. These slots (or memory locations) each have a **unique memory address**. The program counter stores the address of each instruction and tells the CPU in what order they should be carried out.
- When a program is being executed, the CPU performs the fetch-decode-execute cycle, which repeats over and over again until reaching the STOP instruction.

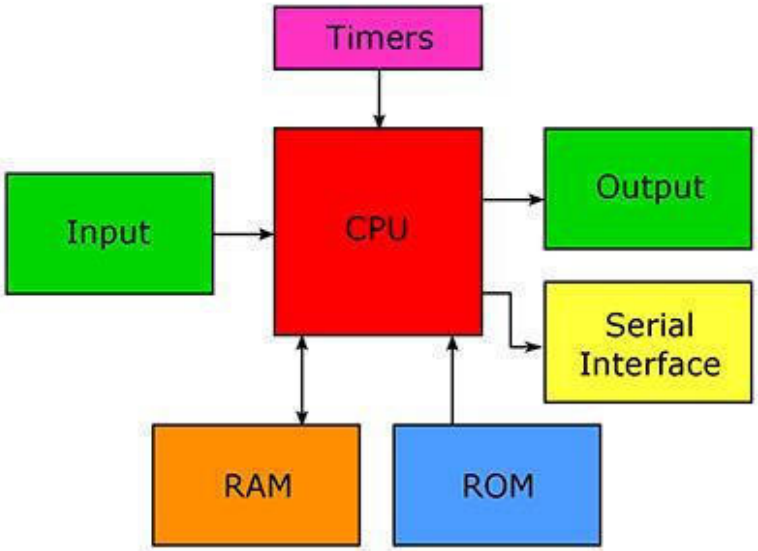
- The processor checks the program counter to see which instruction to run next.
- The program counter gives an address value in the memory of where the next instruction is.
- The processor fetches the instruction value from this memory location.
- Once the instruction has been fetched, it needs to be decoded and executed. For example, this could involve taking one value, putting it into the **ALU**, then taking a different value from a **register** and adding the two together.
- Once this is complete, the processor goes back to the program counter to find the next instruction.

This cycle is repeated until the program ends. We will study more on this……

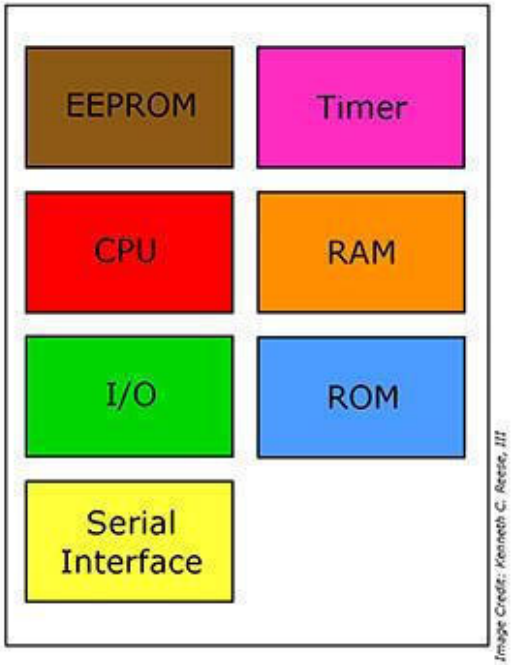


Microprocessor vs microcontroller

Microprocesser:



Microcontroller:



Microprocessor vs. Microcontroller

Microprocessor

- CPU is stand-alone, RAM, ROM, I/O, timer are separate
- designer can decide on the amount of ROM, RAM and I/O ports.
- expensive
- versatility
- general-purpose
- High processing power
- High power consumption
- Instruction sets focus on processing-intensive operations
- Typically 32/64 – bit
- Typically deep pipeline (5-20 stages)

Microcontroller

- CPU, RAM, ROM, I/O and timer are all on a single chip
- fixed amount of on-chip ROM, RAM, I/O ports
- for applications in which cost, power and space are critical
- single-purpose (control-oriented)
- Low processing power
- Low power consumption
- Bit-level operations
- Instruction sets focus on control and bit-level operations
- Typically 8/16 bit
- Typically single-cycle/two-stage pipeline

INTEL CORE i7-3770
SR0PK 3.40GHZ
COSTA RICA
3234B824

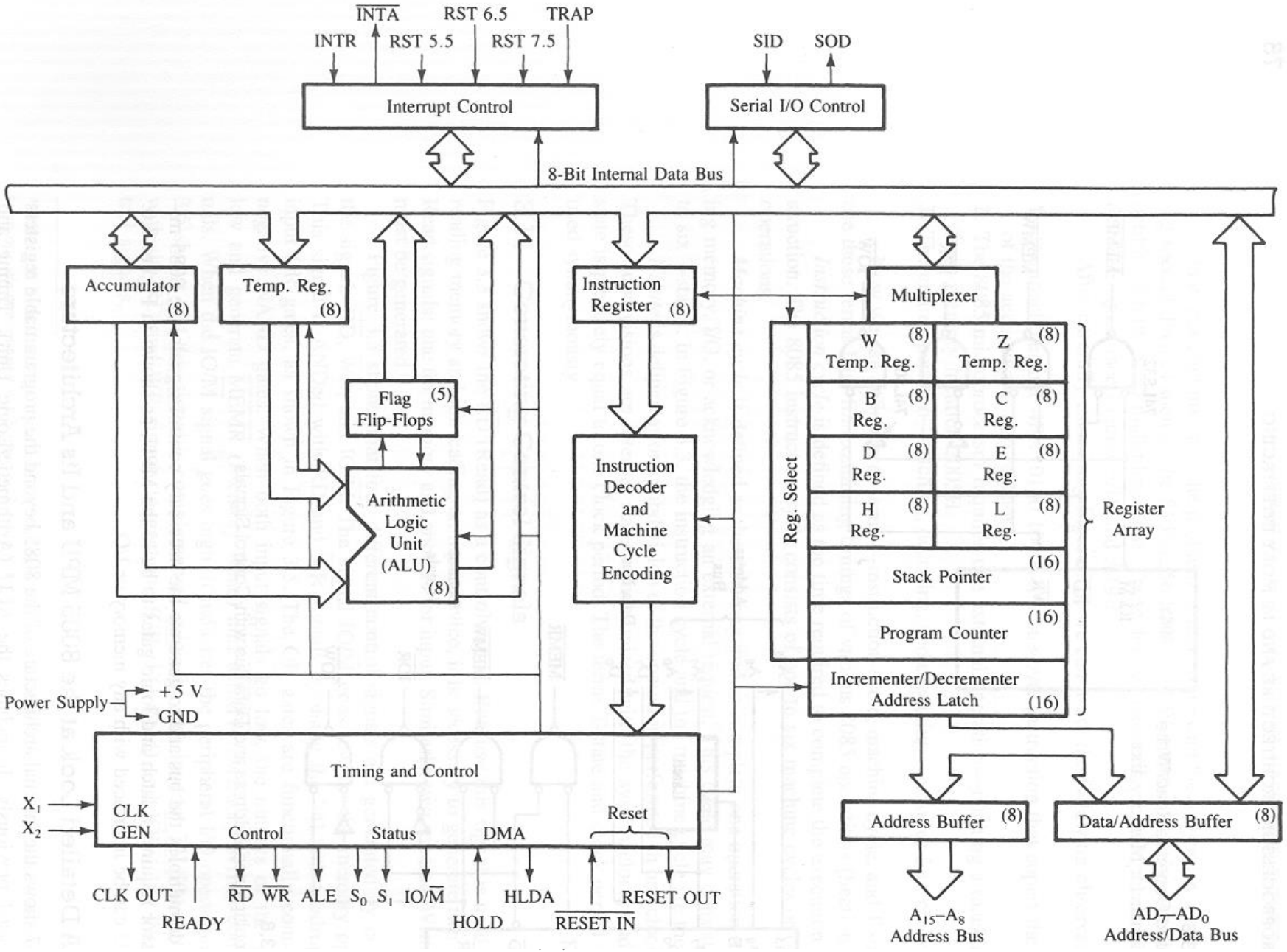
HDMI

History of MPU

NAME	YEAR	TRANSISTORS	DATA WIDTH	CLOCK SPEED
8080	1974	6,000	8 bits	2 MHz
8085	1976	6,500	8 bits	5 MHz
8086	1978	29,000	16 bits	5 MHz
8088	1979	29,000	8 bits	5 MHz
80286	1982	134,000	16 bits	6 MHz
80386	1985	275,000	32 bits	16 MHz
80486	1989	1,200,000	32 bits	25 MHz
PENTIUM	1993	3,100,000	32/64 bits	60 MHz
PENTIUM II	1997	7,500,000	64 bits	233 MHz
PENTIUM III	1999	9,500,000	64 bits	450 MHz
PENTIUM IV	2000	42,000,000	64 bits	1.5 GHz

Year	Architecture	CPU	Core Count	TDP	Clock Speed (MHz)
2004	Prescott	Pentium 570J	1+HT	115	3800
2005	Smithfield	Pentium 840 EE	2+HT	130	3200
2006	Kentsfield	C2E QX6800	4	130	2930
2007	Yorkfield	C2E QX9650	4	130	3000
2008	Nehalem	Core i7-965	4+HT	130	3200 / 3460
2009	Nehalem	Core i7-975	4+HT	130	3360 / 3600
2010	Westmere	Core i7-980X	6+HT	130	3360 / 3600
2011	Sandy Bridge-E	Core i7-3960X	6+HT	130	3300 / 3900
2012	Ivy Bridge	Core i7-3770K	4+HT	77	3500 / 3900
2013	Ivy Bridge	Core i7-4960X	6+HT	130	3600 / 4000
2014	Haswell	Core i7-4790K	4+HT	88	4000 / 4400
2015	Skylake	Core i7-6700K	4+HT	91	4000 / 4200

8085 Microprocessor

- It is an 8-bit microprocessor designed by Intel in 1976 using NMOS technology.
- It has the following configuration –
  - 8-bit data bus
  - 16-bit address bus, which can address upto 64KB
  - A 16-bit program counter
  - A 16-bit stack pointer
  - Six 8-bit registers arranged in pairs: BC, DE, HL
  - Requires +5V supply to operate at 3.2 MHZ single phase clock

8085 Microprocessor internal architecture